



# Vikash Patel

Roll No.:240106078

B.Tech - Biosciences and Bioengineering  
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## EDUCATION

| Degree/Certificate | Institute/Board                         | CGPA/Percentage | Year         |
|--------------------|-----------------------------------------|-----------------|--------------|
| B.Tech             | Indian Institute of Technology Guwahati | 6.36 (Current)  | 2024-Present |
| Senior Secondary   | BSEB Board                              | 75.4%           | 2023         |
| Secondary          | CBSE Board                              | 71.2%           | 2021         |

## PROJECTS

- CVM++ | Stack-Based Virtual Machine and Custom Compiler** Apr. 2026 - May. 2026  
*Personal Project* GitHub
  - Developed a custom programming language and stack-based execution engine in C++, implementing the complete compilation pipeline, from lexical analysis and AST generation to bytecode compilation and runtime execution.
  - Implemented a Lexer and a Recursive Descent Parser to tokenize the source code and construct an Abstract Syntax Tree (AST) supporting variables, arithmetic expressions, comparison operations, conditional statements, and loops.
  - Built a CLI-based REPL and File Runner to compile, inspect, and execute custom language programs, with support for token visualization, AST generation, bytecode inspection, and VM execution.
  - Implemented runtime error handling for invalid syntax, undefined variables, and illegal operations, improving the robustness of the interpreter.
- SecureATM: Banking Management System** Jan. 2026 - Mar. 2026  
*Personal Project* GitHub
  - Built a secure ATM simulation system in C++ using Object-Oriented Programming, modular design, and persistent data storage for account authentication, transaction processing, and record management.
  - Designed and implemented a secure authentication system with features including encrypted PIN management, Masked PIN input, and account verification to protect user credentials.
  - Enhanced runtime efficiency by committing changes to persistent storage only when necessary.
  - Structured the application into 20+ modular functions, separating authentication, transaction processing, file management, and account operations, resulting in a clean, maintainable, and scalable codebase.
  - Designed a secure transaction tracking system that stores transaction details with date, timestamp, and account balance, enabling users to efficiently access a user-specified number of recent transactions.
  - Implemented key banking functionalities including account creation, secure login, PIN management, cash withdrawal, deposit, balance inquiry, transaction history tracking, and profile updates.

## TECHNICAL SKILLS

- Programming** : C/C++, JavaScript \* Elementary proficiency
- Web Development** : HTML, CSS, Tailwind CSS, React.js, Node.js, Express.js
- Database** : MongoDB
- Version Control** : Git, GitHub
- Software and Engineering Tools** : MATLAB, AutoCAD
- Operating Systems** : Windows, Linux
- Core Concepts** : Data Structures and Algorithms, Object-Oriented Programming (OOP), STL, DBMS, Computer Networks

## KEY COURSES TAKEN

- Mathematics** : Differential equation, Complex analysis, Partial Differential Equations
- Computer Science** : Introduction to Computing, Computing Laboratory

## ACHIEVEMENTS

- LeetCode** : Maximum Rating : **1478** || solved **100+** DSA problems || Handle: vikashpatel0207 2026
- Codeforces** : Maximum Rating of : **1170** || solved **50+** CP Problems || Handle: vikash8340 2026

## EXTRACURRICULARS

- Won Silver Medal** in Hockey premier league (Institute's Hockey Tournament) 2025
- Represented my Hostel in Spardha (Inter-hostel sports tournament across 9 Hostels) 2026